



NITTE
(Deemed to be University)

**NMAM INSTITUTE
OF TECHNOLOGY**

(An off-Campus Institution of NITTE (DEEMED TO BE UNIVERSITY), MANGALORE)

Department of MCA

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ADVANCED WEB TECHNOLOGIES LAB ASSIGNMENT 5

1. Student Marksheet Generator

Write a JavaScript program with **HTML form and external JS file** to accept:

Register Number, Student Name ,Marks in 5 subjects using **functions and arrays**, calculate:

Total, Average, Grade, Result (Pass/Fail). Display the result dynamically in the webpage using the **DOM**.

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Marksheet Generator</title>
</head>
<body>

<h2>Student Marksheet</h2>

<form id="form">
  Reg No: <input type="text" id="reg"><br><br>
  Name: <input type="text" id="name"><br><br>
  Marks (5 subjects): <br>
  <input type="number" class="marks"><br>
  <input type="number" class="marks"><br>
  <input type="number" class="marks"><br>
  <input type="number" class="marks"><br>
  <input type="number" class="marks"><br><br>
  <button type="button" onclick="generate()">Generate</button>
</form>
```

```
<h3 id="output"></h3>
```

```
<script src="script.js"></script>
```

```
</body>
```

```
</html>
```

script.js

```
function generate() {  
  let marks = document.querySelectorAll(".marks");  
  let arr = [];  
  
  for (let i = 0; i < marks.length; i++) {  
    arr.push(Number(marks[i].value));  
  }  
  
  let total = arr.reduce((a, b) => a + b, 0);  
  let avg = total / arr.length;  
  
  let grade = (avg >= 90) ? "A" :  
    (avg >= 75) ? "B" :  
    (avg >= 50) ? "C" : "D";  
  
  let result = arr.every(m => m >= 35) ? "Pass" : "Fail";  
  
  document.getElementById("output").innerHTML =  
    `Total: ${total} <br>  
    Average: ${avg} <br>  
    Grade: ${grade} <br>  
    Result: ${result}`;  
}
```

Student Marksheet

Reg No:

Name:

Marks (5 subjects):

81
89
83
92
82

Total: 427

Average: 85.4

Grade: B

Result: Pass

2. Write a JavaScript program to validate a registration form containing: Name, Email, Phone Number, Password. Use **pattern matching (regular expressions)** and **DOM2 event handling** to validate the fields and display suitable error messages.

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Form Validation</title>
</head>
<body>

<h2>Registration Form</h2>

<form id="form">
  Name: <input type="text" id="name"><span id="nErr"></span><br><br>
  Email: <input type="text" id="email"><span id="eErr"></span><br><br>
  Phone: <input type="text" id="phone"><span id="pErr"></span><br><br>
        Password: <input type="password" id="pass"><span
id="passErr"></span><br><br>
  <button type="submit">Submit</button>
</form>

<script src="script.js"></script>
</body>
</html>
```

script.js

```
document.getElementById("form").addEventListener("submit", function(e) {
  e.preventDefault();

  let name = document.getElementById("name").value;
  let email = document.getElementById("email").value;
  let phone = document.getElementById("phone").value;
  let pass = document.getElementById("pass").value;

  let namePattern = /^[A-Za-z ]+$/;
  let emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
  let phonePattern = /^[0-9]{10}$/;
  let passPattern = /^[A-Za-z0-9@#$$%^&+=]{6,}$/;

  document.getElementById("nErr").innerHTML =
    namePattern.test(name) ? "" : " Invalid Name";

  document.getElementById("eErr").innerHTML =
```

```

        emailPattern.test(email) ? "" : " Invalid Email";

        document.getElementById("pErr").innerHTML =
            phonePattern.test(phone) ? "" : " Invalid Phone";

        document.getElementById("passErr").innerHTML =
            passPattern.test(pass) ? "" : " Weak Password";
    });

```

Registration Form

Name:

Email:

Phone:

Password: Weak Password

- Write a JavaScript program to create a **calculator** with buttons for: Addition, Subtraction, Multiplication, Division, Modulus Use **button events** and **textboxes** for input/output also use DOM0 event model.

index.html

```

<!DOCTYPE html>
<html>
<head>
    <title>Calculator</title>
</head>
<body>

<h2>Simple Calculator</h2>

```

```

Enter A: <input type="number" id="a"><br><br>
Enter B: <input type="number" id="b"><br><br>

```

```

<button onclick="add()">+</button>
<button onclick="sub()">-</button>
<button onclick="mul()">*</button>
<button onclick="div()">/</button>
<button onclick="mod()">%</button>

```

```

<h3 id="result"></h3>

```

```
<script src="script.js"></script>
</body>
</html>
```

script.js

```
function getValues() {
    let a = Number(document.getElementById("a").value);
    let b = Number(document.getElementById("b").value);
    return [a, b];
}

function add() {
    let [a, b] = getValues();
    document.getElementById("result").innerHTML = "Result: " + (a + b);
}

function sub() {
    let [a, b] = getValues();
    document.getElementById("result").innerHTML = "Result: " + (a - b);
}

function mul() {
    let [a, b] = getValues();
    document.getElementById("result").innerHTML = "Result: " + (a * b);
}

function div() {
    let [a, b] = getValues();
    document.getElementById("result").innerHTML = "Result: " + (a / b);
}

function mod() {
    let [a, b] = getValues();
    document.getElementById("result").innerHTML = "Result: " + (a % b);
}
```

Output:

+

Simple Calculator

Enter A:

Enter B:

Result: 12

-

Simple Calculator

Enter A:

Enter B:

Result: 8

*

Simple Calculator

Enter A:

Enter B:

Result: 20

/

Simple Calculator

Enter A:

Enter B:

Result: 5

%

Simple Calculator

Enter A:

Enter B:

Result: 0

4. Write a HTML and JavaScript program to accept N elements and store it array (The elements to the array should be added individually when user clicks on the ADD button). Find the sum of even and odd elements and display the result in different textboxes.

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Even Odd Sum</title>
</head>
<body>

<h2>Array Elements</h2>

Enter Number: <input type="number" id="num">
<button onclick="add()">ADD</button><br><br>

Even Sum: <input type="text" id="even" readonly><br><br>
Odd Sum: <input type="text" id="odd" readonly><br><br>

<script src="script.js"></script>
</body>
</html>
```

script.js

```
let arr = [];

function add() {
  let n = Number(document.getElementById("num").value);
  arr.push(n);

  let evenSum = 0, oddSum = 0;
```

```

for (let i = 0; i < arr.length; i++) {
    if (arr[i] % 2 === 0)
        evenSum += arr[i];
    else
        oddSum += arr[i];
}

document.getElementById("even").value = evenSum;
document.getElementById("odd").value = oddSum;

document.getElementById("num").value = "";
}

```

Array Elements

Enter Number:

Even Sum:

Odd Sum:

5. Write a script to input a number n that is the number of the Fibonacci numbers required as output and generate the Fibonacci series accordingly.

index.html

```

<!DOCTYPE html>
<html>
<head>
    <title>Fibonacci Series</title>
</head>
<body>

<h2>Fibonacci Generator</h2>

Enter N: <input type="number" id="n">
<button onclick="generate()">Generate</button>

<h3 id="output"></h3>

<script src="script.js"></script>
</body>

```

</html>

script.js

```
function generate() {  
  let n = Number(document.getElementById("n").value);  
  let a = 0, b = 1, series = "";  
  
  for (let i = 0; i < n; i++) {  
    series += a + " ";  
    let temp = a + b;  
    a = b;  
    b = temp;  
  }  
  
  document.getElementById("output").innerHTML = series;  
}
```

Fibonacci Generator

Enter N:

0 1 1 2 3

6. Write a JavaScript function which receives a number as its parameter and returns the number with its digits in reverse order.

index.html

```
<!DOCTYPE html>  
<html>  
<head>  
  <title>Reverse Number</title>  
</head>  
<body>  
  
<h2>Reverse Number</h2>
```

```
Enter Number: <input type="number" id="num">  
<button onclick="reverseNum()">Reverse</button>
```

```
<h3 id="result"></h3>

<script>
function reverseNum() {
  let num = Number(document.getElementById("num").value);
  let rev = 0;

  while (num > 0) {
    let digit = num % 10;
    rev = rev * 10 + digit;
    num = Math.floor(num / 10);
  }

  document.getElementById("result").innerHTML = "Reversed: " + rev;
}
</script>

</body>
</html>
```

Reverse Number

Enter Number:

Reversed: 54321

7. Write a script that includes a function which receives an array of names as parameter and returns the number of names in the given array that end in either 'ie' or 'y'

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Name Count</title>
</head>
<body>
```

<h2>Count Names Ending with 'ie' or 'y'</h2>

Enter Names (comma separated):

<input type="text" id="names">

<button onclick="check()">Check</button>

<h3 id="result"></h3>

<script>

function countNames(arr) {

let count = 0;

for (let i = 0; i < arr.length; i++) {

let name = arr[i].trim().toLowerCase();

if (name.endsWith("ie") || name.endsWith("y")) {

count++;

}

}

return count;

}

function check() {

let input = document.getElementById("names").value;

let arr = input.split(",");

let res = countNames(arr);

document.getElementById("result").innerHTML =

"Count: " + res;

}

</script>

</body>

</html>

Count Names Ending with 'ie' or 'y'

Enter Names (comma separated):

Emojie, Happy, Hello, Add

Check

Count: 2

8. Write a script that sorts an array with numbers. The order for sorting should be inputted from keyboard either as ascending or descending. Using switch array should be sorted according to inputted order.

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Sort Array</title>
</head>
<body>

<h2>Array Sorting</h2>

Enter Numbers (comma separated):
<input type="text" id="nums"><br><br>

Order (asc/desc):
<input type="text" id="order"><br><br>

<button onclick="sortArray()">Sort</button>

<h3 id="result"></h3>

<script>
function sortArray() {
  let arr = document.getElementById("nums").value
    .split(",").map(Number);

  let order = document.getElementById("order").value.toLowerCase();

  switch(order) {
    case "asc":
      arr.sort((a,b) => a - b);
      break;
    case "desc":
      arr.sort((a,b) => b - a);
      break;
    default:
      document.getElementById("result").innerHTML = "Invalid Order";
      return;
  }

  document.getElementById("result").innerHTML =
    "Sorted: " + arr.join(", ");
}
```

</script>

</body>

</html>

Array Sorting

Enter Numbers (comma separated):

Order (asc/desc):

Sort

Sorted: 8, 12, 20, 34, 50

Array Sorting

Enter Numbers (comma separated):

Order (asc/desc):

Sort

Sorted: 50, 34, 20, 12, 8

Array Sorting

Enter Numbers (comma separated):

Order (asc/desc):

Sort

Invalid Order